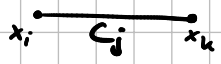
n particles $\quad x = (x_1, x_2, \dots, x_n)^T, \quad v = \dot{x} = (v_1, v_2, \dots, v_n)^T, \quad \ddot{x}$

Constraints $C_j(x) \leftarrow 0$ for all j

Distance constraint: $C_j(x) = \|x_i - x_k\| - l_{ik}$



Constraints handled like linear springs

$x, \Delta x \in \mathbb{R}^i$

1. Pick
   - constraints $C_j(x)$.
   - inverse stiffness α_j
2. Define lambda if $\alpha_j > 0$
3. Calculate $\nabla C_j(x) = \frac{\partial C_j}{\partial x}\Big|_x$
4. Calculate $\tilde{\alpha}_j$
5. Calculate $\Delta\lambda$ using (2)
6. Update x_i that are part of $C_j(x)$ using (1)
7. Update λ_j

$$(1) \quad \Delta x = \underbrace{M^{-1}}_{\mathbb{R}^{i \times i}} \underbrace{\nabla C(x_i)^T}_{\mathbb{R}^{i \times j}} \underbrace{\Delta\lambda}_{\mathbb{R}^{j}}$$

$$(2) \quad \Delta\lambda_j = \frac{-C_j(x_i) - \tilde{\alpha}_j \lambda_{ij}}{\nabla C_j M^{-1} \nabla C_j^T + \tilde{\alpha}_j}$$

$$\tilde{\alpha}_j = \frac{\alpha_j}{\Delta t^2}$$

One λ value per constraint

$$\nabla C(x) = \begin{pmatrix} \text{———} \\ \text{———} \\ \vdots \\ \text{———} \end{pmatrix} \in \mathbb{R}^{j \times i}$$

constraint 0

constraint 1

$\Rightarrow$ Rows correspond to constraints, Cols correspond to particle positions x

Algorithm 1 XPBD simulation loop

```
predict position x̃ ⇐ x^n + Δt v^n + Δt² M⁻¹ f_ext(x^n)

initialize solve x_0 ⇐ x̃
initialize multipliers λ_0 ⇐ 0
while i < solverIterations do
    for all constraints do
        compute Δλ using Eq (18)
        compute Δx using Eq (17)
        update λ_{i+1} ⇐ λ_i + Δλ
        update x_{i+1} ⇐ x_i + Δx
    end for
    i ⇐ i + 1
end while

update positions x^{n+1} ⇐ x_i
update velocities v^{n+1} ⇐ (1/Δt) (x^{n+1} − x^n)
```

Example: $C_j(x) = \|x_i - x_k\| - l_{ik}$

$$\alpha_j = 0.01, \quad \tilde{\alpha}_j = \frac{\alpha_j}{\Delta t^2}$$

$$\frac{\partial C_j}{\partial x_i}\Big|_x = \frac{(x_i - x_k)}{\|x_i - x_k\|}, \quad \frac{\partial C_j}{\partial x_k}\Big|_x = -\frac{(x_i - x_k)}{\|x_i - x_k\|}$$

$$\Delta\lambda_j = \frac{-C_j(x) - \tilde{\alpha}_j \cdot \lambda_j}{m_i^{-1} + m_k^{-1} + \tilde{\alpha}_j}$$

$$x_i \mathrel{+}= m_i^{-1} \cdot \frac{\partial C_j}{\partial x_i}\Big|_x \cdot \Delta\lambda_j$$

$$x_k \mathrel{+}= m_k^{-1} \cdot \frac{\partial C_j}{\partial x_k}\Big|_x \cdot \Delta\lambda_j$$

$$\lambda_j \mathrel{+}= \Delta\lambda_j$$

$$\nabla C_j \cdot M^{-1} \cdot \nabla C_j^T$$

$$= \left(0 \dots 0 \ \frac{(x_i - x_k)}{\|x_i - x_k\|} \ 0 \dots 0 \ -\frac{(x_i - x_k)}{\|x_i - x_k\|} \ 0 \dots 0\right) \cdot \begin{pmatrix} \frac{1}{m_1} & & & & & \\ & \ddots & & & & \\ & & \frac{1}{m_i} & & & \\ & & & \ddots & & \\ & & & & \frac{1}{m_k} & \\ & & & & & \ddots \\ & & & & & & \frac{1}{m_n} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{(x_i - x_k)}{\|x_i - x_k\|} \\ 0 \\ \vdots \\ 0 \\ -\frac{(x_i - x_k)}{\|x_i - x_k\|} \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$= \left(0 \dots 0 \ \frac{(x_i - x_k)}{\|x_i - x_k\|} \ 0 \dots 0 \ -\frac{(x_i - x_k)}{\|x_i - x_k\|} \ 0 \dots 0\right) \cdot \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{1}{m_i} \frac{(x_i - x_k)}{\|x_i - x_k\|} \\ 0 \\ \vdots \\ 0 \\ -\frac{1}{m_k} \frac{(x_i - x_k)}{\|x_i - x_k\|} \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$= \frac{1}{m_i} \underbrace{\left(\frac{(x_i - x_k)}{\|x_i - x_k\|}\right)^2}_{=1} + \frac{1}{m_k} \underbrace{\left(\frac{(x_i - x_k)}{\|x_i - x_k\|}\right)^2}_{=1}$$

$$\overset{v^2 = \|v\|^2}{=} m_i^{-1} + m_k^{-1}$$

$$\frac{\partial c_i}{\partial x_i} = \frac{\partial}{\partial x_i}\left(\|x_i - x_k\| - l_{ik}\right)$$

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

$$= \frac{\partial}{\partial x_i}\|x_i - x_k\|$$

$$= \frac{\partial}{\partial x_i}\sqrt{(x_{i1}-x_{k1})^2 + (x_{i2}-x_{k2})^2 + (x_{i3}-x_{k3})^2}$$

$$= \frac{1}{2\sqrt{(x_{i1}-x_{k1})^2 + (x_{i2}-x_{k2})^2 + (x_{i3}-x_{k3})^2}} \cdot \frac{\partial}{\partial x_i}\left[(x_{i1}-x_{k1})^2 + (x_{i2}-x_{k2})^2 + (x_{i3}-x_{k3})^2\right]$$

$$= \frac{1}{2\|x_i - x_k\|} \cdot \left[\frac{\partial}{\partial x_i}(x_{i1}-x_{k1})^2 + \frac{\partial}{\partial x_i}(x_{i2}-x_{k2})^2 + \frac{\partial}{\partial x_i}(x_{i3}-x_{k3})^2\right]$$

$$= \frac{1}{2\|x_i - x_k\|} \cdot \left[2(x_{i1}-x_{k1}) \cdot \frac{\partial}{\partial x_i} x_{i1} + 2(x_{i2}-x_{k2}) \cdot \frac{\partial}{\partial x_i} x_{i2} + 2(x_{i3}-x_{k3}) \cdot \frac{\partial}{\partial x_i} x_{i3}\right]$$

$$= \frac{1}{\|x_i - x_k\|} \cdot \left[(x_{i1}-x_{k1}) \cdot \begin{pmatrix}1\\0\\0\end{pmatrix} + (x_{i2}-x_{k2}) \cdot \begin{pmatrix}0\\1\\0\end{pmatrix} + (x_{i3}-x_{k3}) \cdot \begin{pmatrix}0\\0\\1\end{pmatrix}\right]$$

$$= \frac{x_i - x_k}{\|x_i - x_k\|}$$

$$\frac{\partial c_i}{\partial x_k} = \frac{\partial}{\partial x_k}\left(\|x_i - x_k\| - l_{ik}\right)$$

$$= \frac{\partial}{\partial x_k}\|x_i - x_k\|$$

$$= \frac{\partial}{\partial x_k}\sqrt{(x_{i1}-x_{k1})^2 + (x_{i2}-x_{k2})^2 + (x_{i3}-x_{k3})^2}$$

$$= \frac{1}{2\sqrt{(x_{i1}-x_{k1})^2 + (x_{i2}-x_{k2})^2 + (x_{i3}-x_{k3})^2}} \cdot \frac{\partial}{\partial x_k}\left[(x_{i1}-x_{k1})^2 + (x_{i2}-x_{k2})^2 + (x_{i3}-x_{k3})^2\right]$$

$$= \frac{1}{2\|x_i - x_k\|} \cdot \left[\frac{\partial}{\partial x_k}(x_{i1}-x_{k1})^2 + \frac{\partial}{\partial x_k}(x_{i2}-x_{k2})^2 + \frac{\partial}{\partial x_k}(x_{i3}-x_{k3})^2\right]$$

$$= \frac{1}{2\|x_i - x_k\|} \cdot \left[2(x_{i1}-x_{k1}) \cdot \frac{\partial}{\partial x_k}(-x_{k1}) + 2(x_{i2}-x_{k2}) \cdot \frac{\partial}{\partial x_k}(-x_{k2}) + 2(x_{i3}-x_{k3}) \cdot \frac{\partial}{\partial x_k}(-x_{k3})\right]$$

$$= \frac{1}{\|x_i - x_k\|} \cdot \left[(x_{i1}-x_{k1}) \cdot \begin{pmatrix}-1\\0\\0\end{pmatrix} + (x_{i2}-x_{k2}) \cdot \begin{pmatrix}0\\-1\\0\end{pmatrix} + (x_{i3}-x_{k3}) \cdot \begin{pmatrix}0\\0\\-1\end{pmatrix}\right]$$

$$= -\frac{x_i - x_k}{\|x_i - x_k\|}$$

Volume Constraint: $C_j(\overset{n \text{ particles}}{p_0, p_1, \dots, p_{n-1}}) = V - V_0$

$$V_{mesh} = \sum_{i=0}^{\overset{\#\text{triangles}}{n_T-1}} \underbrace{\frac{1}{6}(p_{t_{i,0}} \times p_{t_{i,1}})^T p_{t_{i,2}}}_{V_{tetrahedron}}$$

$p_{t_{i,j}}$ is j'th particle of triangle t_i

↳ Closed mesh, can be convex and concave

$$\Rightarrow C_j(p_0, p_1, \dots, p_{n-1}) = \left[\sum_{i=0}^{n_T-1} \frac{1}{6}(p_{t_{i,0}} \times p_{t_{i,1}})^T p_{t_{i,2}}\right] - V_0$$

Derivative: $x = \begin{pmatrix} p_0 \\ p_1 \\ \vdots \\ p_{n-1} \end{pmatrix}$

a, b abh. von x: $\frac{\partial a^T b}{\partial x} \overset{(*)}{=} b^T \frac{\partial a}{\partial x} + a^T \frac{\partial b}{\partial x}$

$\frac{\partial (a \times b)}{\partial x} \overset{(**)}{=} a^\times \frac{\partial b}{\partial x} - b^\times \frac{\partial a}{\partial x}$

↳ $\begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$

$$\frac{\partial C_j(p_0, p_1, \dots, p_{n-1})}{\partial x}$$

$$= \sum_{i=0}^{n_T-1} \frac{1}{6} \frac{(p_{t_{i,0}} \times p_{t_{i,1}})^T p_{t_{i,2}}}{\partial x}$$

$$\overset{(*)}{=} \frac{1}{6} \sum_{i=0}^{n_T-1} \left[p_{t_{i,2}}^T \frac{\partial (p_{t_{i,0}} \times p_{t_{i,1}})}{\partial x} + (p_{t_{i,0}} \times p_{t_{i,1}})^T \frac{\partial p_{t_{i,2}}}{\partial x}\right]$$

$$\overset{(**)}{=} \frac{1}{6} \sum_{i=0}^{n_T-1} \left[p_{t_{i,2}}^T \left(p_{t_{i,0}}^\times \frac{\partial p_{t_{i,1}}}{\partial x} - p_{t_{i,1}}^\times \frac{\partial p_{t_{i,0}}}{\partial x}\right) + (p_{t_{i,0}} \times p_{t_{i,1}})^T \frac{\partial p_{t_{i,2}}}{\partial x}\right]$$

$$= \frac{1}{6} \sum_{i=0}^{n_T-1} \left[p_{t_{i,2}}^T p_{t_{i,0}}^\times \frac{\partial p_{t_{i,1}}}{\partial x} - p_{t_{i,2}}^T p_{t_{i,1}}^\times \frac{\partial p_{t_{i,0}}}{\partial x} + (p_{t_{i,0}} \times p_{t_{i,1}})^T \frac{\partial p_{t_{i,2}}}{\partial x}\right]$$

$x^T y^\times = (x^\times y)^T$

$$= \frac{1}{6} \sum_{i=0}^{n_T-1} \left[(p_{t_{i,2}}^\times p_{t_{i,0}})^T \frac{\partial p_{t_{i,1}}}{\partial x} - (p_{t_{i,2}}^\times p_{t_{i,1}})^T \frac{\partial p_{t_{i,0}}}{\partial x} + (p_{t_{i,0}} \times p_{t_{i,1}})^T \frac{\partial p_{t_{i,2}}}{\partial x}\right]$$

$x^\times y = -y^\times x$

$$= \frac{1}{6} \sum_{i=0}^{n_T-1} \left[(p_{t_{i,2}}^\times p_{t_{i,0}})^T \frac{\partial p_{t_{i,1}}}{\partial x} + (p_{t_{i,1}}^\times p_{t_{i,2}})^T \frac{\partial p_{t_{i,0}}}{\partial x} + (p_{t_{i,0}} \times p_{t_{i,1}})^T \frac{\partial p_{t_{i,2}}}{\partial x}\right]$$

$$= \frac{1}{6} \sum_{i=0}^{n_T-1} \left[(p_{t_{i,1}}^\times p_{t_{i,2}})^T \frac{\partial p_{t_{i,0}}}{\partial x} + (p_{t_{i,2}}^\times p_{t_{i,0}})^T \frac{\partial p_{t_{i,1}}}{\partial x} + (p_{t_{i,0}} \times p_{t_{i,1}})^T \frac{\partial p_{t_{i,2}}}{\partial x}\right]$$

$$\Rightarrow \frac{\partial C_j(p_0, p_1, \dots, p_{n-1})}{\partial p_{t_{i,0}}}$$

$$= \frac{1}{6} \sum_{i=0}^{n_T-1} \left[(p_{t_{i,1}}^\times p_{t_{i,2}})^T \overset{1}{\overset{\parallel}{\frac{\partial p_{t_{i,0}}}{\partial p_{t_{i,0}}}}} + (p_{t_{i,2}}^\times p_{t_{i,0}})^T \overset{0}{\overset{\parallel}{\frac{\partial p_{t_{i,1}}}{\partial p_{t_{i,0}}}}} + (p_{t_{i,0}} \times p_{t_{i,1}})^T \overset{0}{\overset{\parallel}{\frac{\partial p_{t_{i,2}}}{\partial p_{t_{i,0}}}}}\right]$$

$$= \frac{1}{6} \sum_{i=0}^{n_T-1} (p_{t_{i,1}} \times p_{t_{i,2}})^T$$

$$\boxed{\begin{aligned} \Rightarrow \frac{\partial C_j(p_0, p_1, \dots, p_{n-1})}{\partial p_{t_{i,0}}} &= \frac{1}{6} \sum_{i=0}^{n_T-1} (p_{t_{i,1}} \times p_{t_{i,2}})^T \\ \Rightarrow \frac{\partial C_j(p_0, p_1, \dots, p_{n-1})}{\partial p_{t_{i,1}}} &= \frac{1}{6} \sum_{i=0}^{n_T-1} (p_{t_{i,2}} \times p_{t_{i,0}})^T \\ \Rightarrow \frac{\partial C_j(p_0, p_1, \dots, p_{n-1})}{\partial p_{t_{i,2}}} &= \frac{1}{6} \sum_{i=0}^{n_T-1} (p_{t_{i,0}} \times p_{t_{i,1}})^T \end{aligned}}$$

Calculate for every triangle to construct ∇C_j